

UNIT - VII χ^2 - Distribution - Goodness of fit

test:

The test which is used to find how good a fit is there b/w observed frequencies and expected frequencies is called test of goodness of fit.

The χ^2 distribution is used to test how well a fit is there b/w observed frequencies distribution or expected (theoretical) distribution.

The statistic χ^2 is the measure of discrepancy existing b/w observed and expected frequencies.

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - e_i)^2}{e_i}, \text{ where 'k' is no. of cells}$$

(or) classes, ' O_i ' is observed or sample frequencies and ' e_i ' is expected frequency.

χ^2 is a random variable with ' ν ' degree of freedom. The degree of freedom for χ^2 is given by $\text{D.F.} = k - 1$. If the expected frequencies ' e_i ' can be calculated without having to estimate population parameters from the sample

Statistics.

(ii) $v = k - 1 - m$, if ' e_i ' can be calculated by estimating ' m ' parameters from sample statistics.

For poisson, $v = (k - 1) - 1$ ($\because$ ' λ ' is a parameter).

For binomial, $v = (k - 1) - 1$ ($\because$ ' p ' is a parameter).

For normal, $v = (k - 1) - 2$ ($\because$ ' μ ', ' σ ' are parameters).

NOTE:

01) If $\chi^2 = 0$, O_i agrees exactly with e_i .

02) If $\chi^2 > 0$,

(a) χ^2 is small, then O_i and e_i are closer to each other indicating a good fit.

(b) χ^2 is large, then O_i 's are far away from e_i indicating a poor fit.

Conditions for Applying χ^2 -Test:

(i) $N \geq 30$.

(ii) $4 \leq k \leq 16$.

(iii) $\sum O_i = \sum e_i$

(iv) O_i (and e_i) ≥ 10 , otherwise combine the neighbouring frequencies, so that O_i (or e_i) ≥ 10 .